Physics Concepts as Indicative Framework to Model Systems of Economies Ex-ante

Coincise Notes and Conjectures Explorations of Natural Laws as Approach to Preliminarly Structure Economies

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Distllation and exploratory attempt … Interconnection network between movement, force, momentum, energy, and work. Describing systems with Lagrangian and Hamltonian when constraints and starting conditions exist. Modeling systems and optimisation with constraints and conditions. Interwoven system of macro- and microeconomie under constraints, limitations, and conditions.

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WORKING NOTES DRAFT

Potentials

Potential as "intrinsic energy of inertia"

Electro-Magnetic Field (force caused by potential)

- static: lovslized source charge at rest, only spacial dependency, no time dependency
- stationary: source charce at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistc: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movemnt
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := rot(A) \overset{\text{Matter}}{=} \mu_0 \left(\mathcal{H} + \mathcal{M}\right) \overset{\text{Vacuum}}{=} \frac{1}{c}\left(\frac{r}{|r|} \times \mathcal{E}\right) \overset{\text{Far Field}}{=} \frac{1}{2}\frac{1}{2\pi}\mu_0 \frac{k^2 e^{i(kr-\omega t)}}{r}\frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_{0,} \right. \tag{1}$$

Magnetic Field in Matter

Exitation $\hat{=}$ Displacement

$$\text{Exication:} \qquad \mathcal{H} = \frac{1}{\mu_0}\mathcal{B} - \mathcal{M} = \frac{1}{\mu}\mathcal{B} \quad \text{Magnetisation:} \qquad \mathcal{M} = \frac{\mu - \mu_0}{\mu\mu_0}\mathcal{B} = \frac{\mu-\mu_0}{\mu_0}\mathcal{H} \tag{2}$$

Electric Field in Vacuum

$$\mathcal{E} \overset{\text{conservativ}}{=} -\nabla U_e \overset{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t}U_m \overset{\text{fast moving charge}}{=} \frac{1}{2}\frac{1}{2\pi}\frac{1}{\varepsilon_0}\frac{1}{r}\frac{1}{\left(1 - \frac{r}{|r|}\cdot\frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right)\frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} \right.\right.}_{\text{accelerate}} \right. \tag{3}$$

Electric Field in Matter

Displacement $\hat{=}$ Exitation (field change)
dielectricum = non-conducting material
electricum = conductor medium

$$\text{Exication:} \qquad \mathcal{D} = \varepsilon_0\mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0\mathcal{E} + \varepsilon_0\chi\mathcal{E} = \varepsilon_0\left(1 + \chi\right)\mathcal{E} = \quad \varepsilon_0 4\pi r^3 n \left\langle p_m \right\rangle = 4\pi r^3 n \tag{4}$$

Abbreviations

$$\Phi = \text{electric potential} \qquad A = \text{magnetic potential} \qquad j = \nabla \times \mathcal{H} = \text{elec. current density}$$

$$\mathcal{E} = \text{Electric Field} \qquad \mathcal{B} = \text{Magnetic Field} \qquad \nabla\cdot\mathcal{B} = 0 \qquad \mathcal{D} = \text{Displacement density of dielectricum} \qquad \mathcal{M} =$$

$\mathcal{P}$ = Polarisation $\qquad \mathcal{H}$ = magn. exitation, magn. field strength (”magn. displacement”)

n = spacial molecule density in volume

$\alpha = 4\pi r^3 = constant \qquad \chi$ = elec. susceptibility of dieelectricum $\qquad \kappa = \frac{\mu - \mu_0}{\mu_0}$ = magn. susceptibility

$\langle p_m \rangle$ = spacial average of magn. dipol moment $\qquad \langle \mathcal{E}_m \rangle$ = spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field $\qquad \mu$ = magn. permeability $\qquad \mu_0$ = magn. constant ...

ε = dielectricity constant $\qquad \varepsilon_0 = ...$

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation:

...

stationary, slow charge movement:

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t}\mathcal{B} = i\omega\mathcal{B} \qquad \nabla \times \frac{1}{\mu_0}\mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t}\mathcal{E} = j - i\omega\varepsilon_0\mathcal{E} \tag{5}$$

relativistic, fast charge movement:

...

Potential derivation

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}U = J \tag{6}$$

⇒ Potential caused by "ignition J" (inhomgenity) from source at rest, v = 0:

$$U = -\frac{1}{4\pi}\int \frac{J}{|r|}\,dr' \tag{7}$$

elec. Potential structure

$J = -\frac{1}{\varepsilon_0}\varrho$

$$U_e := \Phi \overset{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi}\frac{1}{\varepsilon_0}\int \frac{\varrho}{|r|}\,dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}Q = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}It \tag{8}$$

magn. Potential structure

$J = -\mu_0 j$

$$U_m := A \overset{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi}\int \frac{j}{|r|}\,dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\mu_0 I = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{c^2\varepsilon_0}I = \frac{1}{c^2}\frac{1}{t}U_e \overset{\text{source in uniformal movement},v=const}{=} \tag{9}$$

Electric endogen energy

- Bond Energy
- Binding Energy

$E_{pot} = mc^2$

$$e = \text{electron charge} \qquad q = ne = \text{charge} \qquad n \in \mathbb{N} \tag{10}$$

$$\text{Potential} \qquad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q}mc^4 = \kappa mc^4 \tag{11}$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Exitation Energy

$E_{kin} = \frac{1}{2}mv^2 \mathrel{\hat{=}} (pc)^2 \mathrel{\hat{=}} h\nu = \hbar\omega$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2}\varepsilon_0\mathcal{E}^2 \tag{12}$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \tag{13}$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2}CU^2 \tag{14}$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \tag{15}$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \tag{16}$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda\frac{1}{T} = \sqrt{\frac{k}{m}} \overset{vacuum}{=} kc \overset{matter}{=} k\frac{c}{\eta} = kc\frac{\tan\beta}{\tan\alpha} \overset{\text{electromagnetic}}{=} kc\left(\mathcal{E}_\beta^{-1}\mathcal{E}_\alpha\right)^{-1} = kc\frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc\sqrt{\frac{\varepsilon_0\mu}{\varepsilon\mu}} \tag{17}$$

Phase:

$$\varphi = k\left(\vec{r}\cdot\vec{k}\right) - \omega t := ks - \omega t = ... \tag{18}$$

Phaseshift:

$$\Delta\varphi = \Delta k\left(\vec{r}\cdot\vec{\Delta k}\right) - \Delta\omega t := \Delta ks - \Delta\omega t = ... \tag{19}$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{\lambda_0}{\lambda} = c\sqrt{\varepsilon\mu} \,\hat{=}\, \mathcal{E}_\beta^{-1}\mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = ... \tag{20}$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielctric madium or conductor); Dispersion $\neq$ Dissipation $\neq$ Diffusion

$$... \tag{21}$$

Dielectricity:

$$\varepsilon(\omega) = ... \tag{22}$$

Permeability:

$$\mu(\omega) = ... \tag{23}$$

Conductivity:

$$\sigma(\omega) = ... \tag{24}$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \tag{25}$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c}\frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \tag{26}$$

Energy Balance

$$\frac{\partial}{\partial t}\sum w_i + \cdot \sum \mathcal{S}_{w_i} + j \cdot \mathcal{E} = 0 \tag{27}$$

$$E = E_{kin} + E_{pot} = \sqrt{(pc)^2 + (mc^2)^2} \overset{\text{vacuum}}{=} pc = \hbar\omega$$

Local Force Balance

$$\dot{p} = F \hat{=} \underbrace{\varrho\mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t}\frac{1}{c^2}}_{\text{field force}}\mathcal{S} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}} = 0 \tag{28}$$

Stress Tensor (Tension Force)

Stres Tensor quantifies tension $\hat{=}\Delta$ Forces = Pressure - Traction $\hat{=}$ “torsion tension” at angular momentum

$$\nabla \cdot \mathcal{T} = \varepsilon_0 \left(\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E}\nabla \cdot \mathcal{E}\right) + \frac{1}{\mu_0}\left(\mathcal{B}\nabla \times \mathcal{B} - \mathcal{B}\nabla \cdot \mathcal{B}\right) \tag{29}$$

$$\mathcal{T} \hat{=} \mathcal{T}_{ik} = \delta_{ik}\left(\frac{1}{2}\varepsilon_0\mathcal{E}^2 + \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2\right) - \varepsilon_0\mathcal{E}_i\mathcal{E}_k - \frac{1}{\mu_0}\mathcal{B}_i\mathcal{B}_k \tag{30}$$

$\mathcal{E}$ and $\mathcal{B}$ fields of spacial confined charge and current density distribution, far field, static: $\mathcal{T} \sim \frac{1}{r^4} \Leftrightarrow \int_{\mathbb{R}^3} \mathcal{T}\, d^3r = \int_\infty \mathcal{T}\, df = 0$

Momentum Conservation and Local Tension

linear momentum conservation if $p = const.$, otherwise “tension” occures:

$$p = \int F = \int \frac{d}{dt}p = \sqrt{\frac{1}{c^2}\left(E^2 - (mc)^2\right)} \overset{\text{vacuum}}{=} \frac{E}{c} = \frac{2\pi}{c^2}h\nu$$

$p = const.$, linear, vacuum, Far Field:

$$F = \dot{p} \hat{=} \int\limits_{\mathbb{R}^3} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2}}_{\text{field force density}} \mathcal{S} \right] = \underbrace{\frac{d}{dt} \left[\underbrace{\int\limits_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) +}_{\text{charge momentum}} \underbrace{\frac{1}{c^2}}_{\text{field momentum}} \mathcal{S} \right]}_{= \text{ p momentum conserved if p = const.}} = 0 \tag{31}$$

$p \neq const.$, local $\in \mathbb{R}^2 \ll \mathbb{R}^3 = \infty$, linear, vacuum, Near Field $\Leftrightarrow \exists - \mathcal{T}$ as tension compensation flux:

$$F = \dot{p} \hat{=} \int\limits_{local} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2}}_{\text{field force density}} \mathcal{S} \right] = \underbrace{\frac{d}{dt} \left[\underbrace{\int\limits_{local} (\varrho \mathcal{E} + j \times \mathcal{B}) +}_{\text{charge momentum}} \underbrace{\frac{1}{c^2}}_{\text{field momentum}} \mathcal{S} \right]}_{= p \neq const., \, \exists - \mathcal{T} \text{ local tension}} \overset{local}{=} - \int\limits_{\mathbb{R}^2} \mathbf{n} \cdot \mathcal{T} \; \underbrace{df}_{\text{"skin}} \tag{32}$$

Momentum $p = \int\limits_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S}$

photon Spin $= \int\limits_{\mathbb{R}^3} \frac{1}{c^2} \mathcal{S}$

angular momentum (photon Spin):

$$D = \dot{L} \hat{=} \ldots \tag{33}$$

Abbreviations:

w_i = energy densities (electric w_e, magnetic e_m, gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectros $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Stress Tensor, (tension, momentum flow density)

Mechanical

...

Electromagnectic energy transfer

charge conservation:

$$\frac{\partial}{\partial t}\varrho + \nabla \cdot j = 0 \tag{34}$$

$\varrho = \sigma\delta =$ charge density in interior

$\sigma =$ charge density at border (skin)

$\delta =$ Dirac delta-function

$j = \sigma\left(\mathcal{E} + v \times \mathcal{B}\right) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t}\mathcal{D} \hat{=}$ elecectric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity $= \left|\bar{\mathcal{S}}\right|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t}\left(w_e + w_m\right) + \ \cdot \mathcal{S} + j \cdot \mathcal{E}\right) dr = \frac{d}{dt}W_e + \frac{d}{dt}W_m + D_S + \int \left(j \cdot \mathcal{E}\right) dr \overset{\text{low frequency}}{=} \frac{1}{2}\frac{d}{dt}\left(LI^2\right) + \frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right) \tag{35}$$

Abbreviations:

$\mathbf{n} =$ surface normal vector (orthongoal on skin)

$\Delta \cdot :=$ div grad $=$ Laplacian

$J =$ "ignition of inhomogenity,cause"

$j =$ Current density

$\varrho =$ Charge density

$V =$ Potentialdifference

$I =$ Current

$R =$ Resistance

$L =$ Inductance

$C =$ Capacitance

$Q =$ charge (ion)

$\Psi = LI =$ magn. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector, radiation intensity

$D_S = \int (\cdot \mathcal{S})\, dr = \int \operatorname{div}(\mathcal{E} \times \mathcal{H})\, dr$ = dissipation engergy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}}$ = skin depth (surface) of conductor

$r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho$ = radius (extension) of conductor

$\rho = kr$ = "radius of curvature k"

μ = permeability of conductor

σ = conductivity of conductor (dielectricum)

Thermal

...

Gravitational

...

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \tag{36}$$

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \tag{37}$$
$$L \approx const. \qquad C \approx const. \qquad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \tag{38}$$

$$L(t) \neq const. \qquad C(t) \neq const. \qquad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electic Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emgergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$c = \frac{1}{\sqrt{\varepsilon_0\mu_0}} = \nu\lambda = \frac{\lambda}{T}$
$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \qquad \nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

Refraction index η emerges: $\varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \mathrel{\widehat{=}} \left(\frac{\eta}{c}\right)^2$
Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

static, vacuum, granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

Far Field

static, vacuum, flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2} = 1$$

Dissipation in Vacuum

no Contiunuity of chrge/matter, no Refraction index, Reflection

Dissipation in Matter

no Contiunuity of charge/matter with dielectricum, refraction merges, reflection, absorbtion

Energy gain by excitation

Action Quantum, Planck Constant (Energy-Frequency-Slope)[1]:

[1] Wirkungsquantum

$$
\begin{aligned}
h = \tan(\varphi) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\
\frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda
\end{aligned}
\tag{39}
$$

$$
\hbar = \frac{E_1 - E_2}{\omega} = \tag{40}
$$

$$
E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \tag{41}
$$

$$
V_0 = \text{Opposing Potential } (I = 0)
$$

$$
\varphi = \angle(E_{kin}, \nu) \qquad e = \text{electron charge} \qquad Q = \text{ion charge}
$$

$$
C = \text{Capacitance} \qquad \nu = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}\sqrt{1 - \frac{C}{L}R^2}
$$

$$
\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C}\left(\frac{e}{h}Q + \nu_0\right)^2
$$

Ionisation Energy of chemical element $E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{h}{k\lambda}\omega$

Energy released (gained), see... Equation 6

- with mass (e.g. electron), Tunnel-Effect: $E_{gain} = E_{pot} - E_i = mc^2 - h\nu_i$
- massless (e.g. photon), Doppler-Effect (frequency shift): $E_{gain} = E_{kin} - E_i = h\left(\nu_0 - \nu_i\right) = \Delta\omega\hbar$

Frequency Cause-Effect

Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta k v = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \overset{electromagnetic}{=} \dots = \sqrt{\frac{1}{LC}} \overset{QM}{=} \frac{1}{2}\frac{p^2}{m}\frac{1}{\hbar} \tag{42}$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v}\frac{1}{\lambda} = \frac{\omega}{k}\frac{1}{\lambda} = \frac{\omega}{2\pi} \tag{43}$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{\hbar}\omega\mathcal{T} + \nu_0 \tag{44}$$

$$e = \text{electron charge} \qquad V_0 = \text{Opposing Potential } (I = 0) \tag{45}$$

Energy Level: $i = [1, n] \in \mathbb{N}$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \tag{46}$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurance of an event/state by space distance and time periodicity as well as movement speed and reach length, $v^2 = v_{ph}v_{gr}$, $\omega t = 2\pi\nu t = \lambda \neq 0$:

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \tag{47}$$

Rest state of source charge

Statics localization of charge. Only spacial demependeny, no time dependency.

Energy transport ... Information transmission ...

$v = 0 \qquad \dot{v} = 0$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v = const. \ll c \qquad \dot{v} = 0$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v \approx c \qquad \dot{v} = 0$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v \neq 0 \qquad \dot{v} \neq 0$

Movement in Vacuum

$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{c}{v_{Ph}} = 1 \qquad 2\pi = k\lambda \qquad v_{ph} = c \qquad \omega = kc$

$\varepsilon_0 \mu_0 = \frac{1}{c^2}$

Continuity: $\nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$

Movement in Matter

$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{c}{v_{Ph}} = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \neq 1 \qquad v_{ph} = \pm\sqrt{\frac{1}{\varepsilon\mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \qquad \omega = kv_{ph} = kc\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} = kc\frac{\tan\beta}{\tan\alpha} = k\frac{c}{\eta}$

$\varepsilon\mu = \left(\frac{\eta}{c}\right)^2$

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

$r_0 \ll r \ll \lambda \qquad kr \ll 2\pi \qquad \frac{w_m}{w_e} \ll 1$

r is small, fast movement

Far Field

$r \gg \lambda$ $\quad kr \gg 1$ $\quad r \gg kr_0^2 \ll 1$ $\quad \frac{w_m}{w_e} = 1$

r is big, slow movement

High Frequency

$\omega \gg \frac{2}{\mu \sigma r_0^2}$

λ is big, ν frequency i small, short time period

Low Frequency

$\omega \ll \frac{2}{\mu \sigma r_0^2}$

λ is small, ν frequency i big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global: ....

Local: ....

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1}\mathcal{D} = \mathcal{E}^{-1}\left(\mathcal{P} + \varepsilon_0 \mathcal{B}\right)$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1}\mathcal{H} = \mathcal{B}^{-1}\left(\frac{1}{\mu_0}\mathcal{B} - \mathcal{M}\right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1+\omega^2 t^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2}\varepsilon_0 \mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta \frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan\alpha}{\tan\beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$

Conditions (skin, interface, border, surface)

Boundary Conditions

$\mathbf{n} \cdot [\mathcal{D}] = \sigma$

$\mathbf{n} \cdot [\mathcal{B}] = 0$

$\mathbf{n} \times [\mathcal{E}] = 0$

$\mathbf{n} \times [\mathcal{H}] = J$

Interior Conditions

e.g. for Superconductivity

$\mathbf{n} \cdot \mathcal{H} = 0$

$\mathbf{n} \times \mathcal{D} = 0$

Abreviations

$\mathbf{n} =$ normal vector (orthogonality orientation)

$\mathcal{E} =$ elec. Field ("traction force"?)

$\mathcal{B} =$ magn. Field ("compressive force"?)

$\mathcal{D} =$ elec. Displacement Field ("exitioan")

$\mathcal{H} =$ magn. Exitation Field ("incentivation")

$\sigma =$ electric charge at surface

$J =$ current density at surface

Distinguishing Situations Conditions and Dependencies

effects of charge (separaton) and differentieting situations (conditioning frameworks) for fields and potentials:

Spacetime Embeding

- Spacial dependence (mechanics)
- Time dependence (dynamics)

Movement Framework

- Static: rest system, equilibrium
- Dynamic:
  - Stationary: slow motion
  - Relativistic: fast motion
  - Radiation; accelerated motion

Medium Property

- Vacuum: spacial mean = time mean
- Matter:
  - Mass (inertia)
  - Charge (conductor, ion)
  - Dielectricum (isolator)
  - Dispersion (refraction)

Field Distance

- Near Field (granular spacetime, mean average values)
- Far Field (flat spacetime)

Localisation Identity

- Global (regular)
- Local (singular)

-> Tool concepts: Momentum, Force, Energy

-> Conditions framework: - continuity condition (charge, current) - equilibrium conditions (action vs. reactio, homgenity vs. inhomogenity) - initial conditions (starting point) - border (skin) conditions (differentiation) - interior conditions (integration) - conservations of local densities (charge, energy, momentum)

-> Properties: - characteristics from charge: dielectricity ε, permeability μ, conductivity σ, frequency ω - characteristics from heat: … - Properties frm gravitation: …

Operators

$\nabla := grad$ = Gradient (Gefälle, Neigung)

$\nabla\cdot := div$ = Divergence (Quelle)

$\nabla\times := rot$ = Rotation (Wirbelung, Curl)

∇^2 = Lagrange Operatonr (Krümmung, Beschleunigung)

$\Delta\cdot := div(grad)$ = Laplace Operatorn

$\Delta :=$ Difference, shift

Acronyms

Acronyms of Physics: ?@sec-workenergy

A = Action	$E = E_{kin} + E_{pot}$ = total energy
t = time	r = dpace (distance between two points, one-dimensional length)
v = velocity	c = light speed
p = momentum	F = force
I = inertia	$\mathcal{L}$ = Lagrangian
$\mathcal{H}$ = Hamiltonian	$\mathcal{K}$ = kinetic Energy
$\mathcal{U}$ = effective potential Energy ($\in E_{pot}$)	$\mathcal{V}$ = potential Energy ($\in E_{pot}$)
$\mathcal{Z} = \frac{1}{2}\frac{L_0^2}{mr^2}$ = Centrifugal Potential	$V = r_1 \cdot r_2 \cdot r_3$ = Volume
k = Wave Vector ("curvature")	W = Work (done vs. received)
P = Pressure	L_0 = angular momentum
T = endogen Temperature	H = exogen Heat
U = endogen Energy ($E_{kin} + E_{pot}$)	$\Phi = \frac{\mathcal{V}}{q}$ = Electric Potential
$\mathcal{A}$ = Magnetic Potential	b_0 = Boltzmann constant
g_0 = Gas constant	m = mass
q = charge	\$$n$ = amount of objectes (particles density $n = \frac{N}{V}$), $n = 2$ and $f = 1$
ϵ_0 = electric constant	$f = 3n \pm z$ = degrees of freedom, $f = 1$ and $n = 2$
μ_0 = magnetic constant	$\mathcal{B}$ = Magnetic Field
$\mathcal{E}$ = Electric Field	H$\mathcal{C} = m \cdot c_0 = \frac{\Delta H}{\Delta T}$ = heat capacity
$c_0 = \frac{1}{m}\frac{\Delta H}{\Delta T}$ = specific heat	$S = b_0 \cdot \ln(\Omega)$ = Entropy (macro state)

$l =$ Moll quantity	$\Omega =$ micro states
$z =$ amount of constraints (boundry conditions)	$b_0 =$ Boltzmann constant
$\kappa = \frac{c_P}{c_V} = \frac{f+2}{f} =$ adiabaty	$\iota = \kappa$ adiabat
$\iota = 0$ isobar	$\iota = 1$ isotherm
$\iota = \infty$ isochor	...
...	...

Acronyms of Economy: ?@sec-productivityvalue

$T =$ Taxes	$M =$ Import of Goods and Services from foreing symstes
$G =$ Governent Expenses, incl. Social Insurances	$X =$ Export of Goods and Services to foreign system
$Y =$ Income of Economy from Turnover	$G_A =$ Governental Subsidies
$D_A =$ Depreciations (Reinvestments) on Assets	$V_N =$ Net Naötional Production, Society NNP
$N =$ Monetary Quantity	$Q =$ Monetary Turnover Velocity
$V_I =$ Gross Domestic Product GDP $= \frac{Output}{Input}$, Tradevolume	$P =$ Price niveau (Inflation adjusted Value)
$L =$ Wages from Labor Work (Salaries, ...)	$R =$ Returns, Earnings, Gains
$Y_A =$ Income of priv. Business Households (Companies, Services, Real Estate Rentals, Retained Profits)	$Y_H =$ Income from priv. Capital Households (Interests, Coupons, Dividends, ... of priv. Assets, Investmens, Credits, Debits, Bonds, Equity)
$T_A =$ Tax on Capital of Corprate Compaies (Business Assets)	$Y_G =$ Governmental Income from Assets, Services, Social Institutions/Insurances
$Z_G =$ Interests on Governmental Debt	$V_S =$ Gross National Produkt, Society GNP
$I =$ Investments on Assets, incl. Storage Change	$R_M =$ Capital Earnings and Wages from Abroad (from Foreign System)
R_X Capital Earnings and Wages to Abroad (to Foreign System)	$W =$ Expensens, costs from human and machinary work efforts

see also [1]

1. Wachter A, Hoeber H (2005) Compendium of Theoretical Physics. Springer